Probabilidad	**Fórmulas** **Corte 3**	**Profesora:** **Diana Carolina** **Moreno Chavarro**

Distribución **Distribución**	**Función** **de probabilidad**	**Valor** **esperado**	**Varianza**	**FGM**
Binomial	$P(Y=y)=\binom{n}{y}p^y(1-p)^{n-y}$	np	npq	$[pe^t+(1-p)]^n$
Geométrica	$P(Y=y)=q^{y-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Binomial Negativa	$P(Y=y)=\binom{y-1}{r-1}p^rq^{y-r}$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$	$\left(\frac{pe^t}{1-(1-p)e^t}\right)^r$
Hipergeométrica	$P(Y=y)=\frac{\binom{r}{y}\binom{N-r}{n-y}}{\binom{N}{n}}$	$\frac{nr}{N}$	$n\left(\frac{r}{N}\right)\left(\frac{N-r}{N}\right)\left(\frac{N-n}{N-1}\right)$	
Poisson	$P(Y=y)=\frac{\lambda^y e^{-\lambda}}{y!}$	λ	λ	$e^{\lambda(e^t-1)}$
Uniforme	$f(y)=\begin{cases}\frac{1}{b-a} & si \quad a\le y\le b\\ 0 & en\ otro\ caso\end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Normal	$f(y)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right\}$	μ	σ^2	$exp(\mu t+\frac{t^2\sigma^2}{2})$
Gamma	$f(y)=\begin{cases}\frac{y^{\alpha-1}e^{-y/\beta}}{\beta^\alpha\Gamma(\alpha)} & 0\le y\le\infty,\\ 0 & en\ otro\ caso.\end{cases}$	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
Exponencial	$f(y)=\begin{cases}\frac{1}{\beta}e^{-y/\beta}, & y>0\\ 0, & e.o.c.\end{cases}$	β	β^2	$(1-\beta t)^{-1}$

Cuadro 1: Distribuciones de probabilidad